

# HARIRAMAKRRISHNAN RAMACHANDRAN

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## PROFESSIONAL EXPERIENCE

**SOFTWARE ENGINEER – HCL Technologies | Chennai, India**

**09/2021 – 01/2025 (3.5 Years)**

- Built and fine-tuned **NLP and ML models** using Python (scikit-learn, PyTorch, Hugging Face Transformers) for classification, entity extraction, and text generation tasks.
- Designed **end-to-end ML training and inference pipelines** with data preprocessing, feature engineering, model evaluation, and versioning using MLflow.
- Developed **LLM-powered automation scripts** using OpenAI and Hugging Face APIs, reducing manual document processing time by 40% for internal operations teams.
- Built **REST APIs with FastAPI and Flask** to serve ML model predictions as real-time microservices, integrated into Java Spring Boot-based backend systems.
- Worked with **PostgreSQL and AWS S3** to manage training datasets, feature stores, and model artefacts at scale, ensuring data lineage and reproducibility.
- Collaborated with product and data engineering teams in **Agile/Scrum** sprints to define ML problem framing, evaluation metrics, and production deployment criteria.
- Automated model retraining and evaluation workflows using **Jenkins and Python**, ensuring continuously updated models without manual pipeline intervention.
- Applied **prompt engineering and RAG (Retrieval Augmented Generation)** patterns to ground LLM outputs in domain-specific knowledge bases, improving answer accuracy by 35%.

## SKILLS

**Machine Learning & AI** – scikit-learn, PyTorch, Hugging Face Transformers, LLM APIs, RAG, NLP.

**Programming Languages** – Python (advanced), SQL, Java, Bash scripting.

**Python Ecosystem** – pandas, NumPy, Flask, FastAPI, LangChain, MLflow.

**Databases & Storage** – PostgreSQL, MySQL, AWS S3, vector databases (Pinecone basics).

**Cloud & MLOps** – AWS (EC2, S3, Lambda, SageMaker basics), Docker, Jenkins, CI/CD pipelines.

**Development Practices** – Agile/Scrum, Git, experiment tracking, model versioning, prompt engineering.

**Tools** – Jupyter Notebook, VS Code, Postman, Jira.

## EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of  
Ireland, Dublin**

**02/2026**

**BACHELOR OF ENGINEERING in Computer Science – SNS College  
of Technology, Coimbatore, India**

**01/2021**

## PROJECTS

**LLM-Powered Document Classification & Extraction (HCL Technologies)**

- Built an NLP pipeline using Hugging Face Transformers to classify and extract structured data from unstructured support documents, reducing manual classification effort by 40%.
- Applied prompt engineering with OpenAI API to generate structured summaries from long-form technical reports, improving information retrieval speed for operations teams.
- Deployed the pipeline as a FastAPI microservice integrated into the existing Java Spring Boot backend, enabling real-time document processing.

#### **ML-Driven Anomaly Detection for Operational Monitoring (HCL Technologies)**

- Developed an anomaly detection system using scikit-learn (Isolation Forest, LSTM) to identify system metric deviations from historical baselines, reducing alert noise by 35%.
- Automated model retraining with Jenkins pipelines and stored artefacts in AWS S3, ensuring continuously updated detection thresholds without manual intervention.

### **ACHIEVEMENTS & CERTIFICATIONS**

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)